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Strict proof and applicable range of the quarter-wavelength model for microwave absorbers

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Abstract

In the research and applications on microwave absorption materials and electromagnetic absorbers, the quarter-wavelength model is widely used in describing the dependence of the reflection loss (RL) peak frequency on electromagnetic parameters and thickness, and in the preparation and design of microwave absorption materials. However, strict proof regarding this model based on transmission line theory is lacking. Moreover, a deviation between the absorption peaks obtained from the RL results and those from the quarter-wavelength model is observed in some materials, for which a universal and reasonable theoretical explanation is also lacking. Hence, in this work, we derive strict proof of the quarter-wavelength model based on transmission line theory, and develop a revised equation for this model to explain the above-mentioned deviation. The revised equation could well fit the deviations observed in our experimental results. Combined with the experimental results, we also discuss the applicable ranges for the quarter-wavelength model and the revised equation. Moreover, the results of this work show that the appropriate value of the phase angle is effective in reducing the absorber thickness.

Keywords: microwave absorption peak, quarter-wavelength model, transmission line theory, microwave absorption materials

(Some figures may appear in colour only in the online journal)

1. Introduction

With the increasing need for electromagnetic absorbers in both military and civilian applications, microwave absorption materials have attracted considerable attention [1, 2, 3–7]. The basic requirement of a microwave absorber is that when the microwaves of a specific frequency range are incident on the surface of the absorber, the proportion of the reflected waves detected should be extremely low [3, 8, 9]. In the reflection loss (RL)—frequency (f) curve, which is generally used to describe microwave absorption properties, this requirement is based on the fact that there is a strong minimum in RL (called the RL peak) in the desired frequency range [9]. However, the RL–f curve includes not only cases with RL peaks (showing good absorption performance), but also those without RL peaks (showing poor absorption performance) [10–12]. It is difficult to directly specify the condition for the existence of absorption peaks using the RL curve [10, 13]. Therefore, considering practical applications, it is essential to establish an absorption peak model to simply clarify the relationship between the RL peak frequency and complex permeability ($\mu_r = \mu' - j\mu''$), permittivity ($\varepsilon_r = \varepsilon' - j\varepsilon''$) and absorber thickness (d) [9].

Currently, the quarter-wavelength model, $d = \frac{(2k+1)c}{4f\sqrt{|\mu_r\varepsilon_r|}}(k = 0, 1, 2, \ldots)$, is more applicable and has fewer limitations

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compared to the other models for explaining the RL peak frequency [14, 15]. For example, the widely-used model proposed by Naito and Suetake [16] completely fails in explaining the absorption peaks exhibited by dielectric materials and is unable to predict the peak frequency and absorption intensity of magnetic materials quantitatively [9, 11, 17, 18]. In contrast, the quarter-wavelength model can describe the dependence of the RL peak frequency on electromagnetic parameters and thickness in most microwave absorption materials, especially materials with high permeability or permittivity [1, 6, 8, 12, 18–21]. However, this model is a semi-empirical formula. Since it was proposed in the 1990s, the model has always been understood and interpreted from the viewpoint of the destructive interference of electromagnetic waves [9, 14, 22, 23]. Strict proof of the model based on transmission line theory is still lacking. Hence, there is limited understanding about this model.

Moreover, in some materials, a deviation between the peak–thickness results (the RL peak frequencies and their corresponding thicknesses) obtained from the RL curves and those fitted by the quarter-wavelength model is observed [9, 24–26]. There are no universal and reasonable theoretical explanations for this deviation. For example, in the reduced graphene oxide/carbonyl iron powder/polyvinyl alcohol composites reported by Zhang's group, the peak–thickness results obtained from the RL curves deviate from the fitting curves based on the quarter-wavelength model [26]. In the magnetite–thermoplastic natural rubber nano-composite showing two RL peaks in the 1–20 GHz range for a certain thickness, the peak frequency and absorber thickness corresponding to the first RL peak at a low frequency do not satisfy the quarter-wavelength formula [24]. Kong's group attributed this RL peak to magnetic loss ($|\mu_r| = |\varepsilon_r|$) [24]. This deviation was also observed in the RL curves of our previously reported BN/GO/Fe_3O_4 composite, and the concept of complex thickness was introduced to explain the dependence of the RL peak frequency on electromagnetic parameters and absorber thickness [25].

Therefore, our objective was to clarify the above problems regarding wave absorption mechanisms in this work. Based on transmission line theory, considering all the RL peaks and the minimum condition of the RL–f curves, we obtained an exact solution to the matching condition of peak frequencies for microwave absorbers backed with perfect conductors. On this basis, the quarter-wavelength model and a revised equation addressing the above-mentioned deviation were derived. Furthermore, combined with the experimental results, the applicable ranges for the quarter-wavelength model and the revised equation were identified.

2. Derivation of quarter-wavelength formulas

2.1. Equivalent transformations of related equations

Based on transmission line theory [27], when plane waves are normally incident on the surface of a single-layer absorber (μ_r, ε_r) backed by a perfect conductor, considering the continuity of the vacuumabsorber interface, the RL–f curve at a given absorber thickness (d) can be calculated from μ_r and ε_r using the following expressions [1, 14]:

$$RL(\mathrm{dB}) = 20\log\left|\frac{Z_p - Z_1}{Z_p + Z_1}\right| = 20\log|\Gamma| \tag{1}$$

$$Z_P = Z_2\tanh\left(j\frac{2\pi f}{c}\sqrt{\mu_r\varepsilon_r}d\right). \tag{2}$$

Here, Z_p is the input impedance (refers to $\vec{E}/\vec{H}$ of the plane wave on the inner side of the vacuumabsorber interface); Γ is the reflection coefficient; $Z_1 = \sqrt{\mu_{r1}/\varepsilon_{r1}} = 1$; $Z_2 = \sqrt{\mu_r/\varepsilon_r}$; f is the frequency of the electromagnetic wave.

Further, μ_r and ε_r were converted into their exponential forms: $\mu_r = |\mu_r|e^{-j\delta_m}$ and $\varepsilon_r = |\varepsilon_r|e^{-j\delta_e}$, where $\tan\delta_m = \mu''/\mu'$ and $\tan\delta_e = \varepsilon''/\varepsilon'$. Then, we define two phase angles as follows:

$$\psi_1 = \frac{1}{2}(\delta_m - \delta_e) \tag{3a}$$

$$\psi_2 = \frac{1}{2}(\delta_m + \delta_e). \tag{3b}$$

ψ_1 in equation (3*a*) is the phase angle of $\sqrt{\mu_r/\varepsilon_r}$. According to the relationship ($\vec{E} = \vec{u}\times\mu_0\mu_r\vec{H}$) between the electric intensity vector ($\vec{E}$) and magnetization vector ($\vec{H}$) of the plane wave, ψ_1 indicates the phase difference between $\vec{E}$ and $\vec{H}$ caused by the medium itself ($\sqrt{\mu_r/\varepsilon_r}$). In addition, ψ_2 in equation (3*b*) is the phase angle of $\sqrt{\mu_r\varepsilon_r}$, related to the phase difference caused by propagation in the medium.

Similarly, $\tanh(j\frac{2\pi f}{c}\sqrt{\mu_r\varepsilon_r}d)$ in equation (2) was also converted into the following exponential form: $A_de^{-j\theta}$. Hence, Z_p in equation (2) can be written as:

$$Z_p = \sqrt{\left|\frac{\mu_r}{\varepsilon_r}\right|}A_de^{-j(\psi_1+\theta)} = Ae^{-j\psi} \tag{4a}$$

$$A = \sqrt{\left|\frac{\mu_r}{\varepsilon_r}\right|}A_d = \sqrt{\left|\frac{\mu_r}{\varepsilon_r}\right|}\frac{1}{2\cos\theta}\frac{\sinh(2\alpha d)}{\cos^2(\beta d)+\sinh^2(\alpha d)} \tag{4b}$$

$$\tan\theta = \frac{\sin(-2\beta d)}{\sinh(2\alpha d)} \tag{4c}$$

where α (β) is the real (imaginary) part of $\gamma = j\frac{2\pi f}{c}\sqrt{\mu_r\varepsilon_r}$. A is the amplitude of Z_p, referring to the amplitude ratio $\vec{E}/\vec{H}$ of the plane wave on the inner side of the vacuum–absorber interface. $\psi = \psi_1 + \theta$ is the phase angle of Z_p, referring to the phase difference between $\vec{E}$ and $\vec{H}$ on the inner side of the vacuum–absorber interface. Substituting $Z_p = Ae^{-j\psi}$ into $|\Gamma|$ defined in equation (2), we obtained:

$$|\Gamma| = \left|\frac{Z_p - 1}{Z_p + 1}\right| = \sqrt{2\frac{A^2+1}{A^2+1+2A\cos\psi} - 1}. \tag{5}$$

2.2. Phase matching condition of RL peaks

As the RL (dB)–f curves show, the functional relationship between RL and f, the RL peak frequencies are located at the minimum points of the RL–f curve. According to equation (1), the minimum points of RL are equivalent to the minimum points of $|\Gamma|$, corresponding to $|\Gamma|_f{}' = 0$ and $|\Gamma|_f{}'' > 0$. According to the derivation shown in the appendix, this condition is equivalent to $|\Gamma|_\psi{}' = 0$ and $|\Gamma|_\psi{}'' > 0$.

In equation (5), there is another variable A which changes with μ_r, ε_r and frequency for a given thickness. However, on the one hand, the influence of the change in μ_r (ε_r) with f is ruled out in the differential equations to discuss universal results, as μ_r (ε_r) changes differently with f in different absorbers. On the other hand, as mentioned above, A refers to the amplitude ratio $\vec{E}/\vec{H}$ of the plane wave on the inner side of the vacuumabsorber interface; so, A changes little around each RL peak frequency according to its physical significance. This is supported by the observation that A changes slowly and monotonically with f in the application frequency range in the absorption materials. Hence, we assumed that A remained unchanged near peak frequencies in the following differentiations. Then, the derivatives of $|\Gamma|$ versus ψ yielded

$$\psi = \psi_1 + \theta = 2n\pi. \tag{6}$$

Therefore, the phase angle of Z_p should meet the condition $\psi = 2n\pi$ $(e^{-j\psi} = 1)$ if an RL peak appears. We termed this condition the phase matching condition.

2.3. Dependence of RL frequency on electromagnetic parameters and thickness

2.3.1. $\psi_1 = 0$. Generally, in microwave-absorbing materials, the phase difference between $\vec{E}$ and $\vec{H}$ caused by $\sqrt{\mu_r/\varepsilon_r}$ can be ignored. We considered $\psi_1 = 0$ these materials. Hence, the phase matching condition can be written as $\psi = \theta = 2n\pi$. In turn, according to equation (4*c*), we obtained $2\beta d = K\pi$.

Moreover, because the log function is monotonously increased, a smaller value of $|\Gamma|$ will lead to a stronger RL peak. Theoretically, as the so-called perfect matching points out, the strongest peak appears at $Z_p = 1$ [10, 15]. Hence, when the phase matching condition is satisfied, the value of A should be close to 1 to obtain prominent RL peaks. Clearly, $\sqrt{|\mu_r/\varepsilon_r|} < 1$, the value of A_d should be larger than 1 according to equation (4*b*). Therefore, K is an odd number $(K = 2k+1)$, which means $2\beta d = (2k+1)\pi$. As mentioned above, β is the imaginary part of $j\frac{2\pi f}{c}\sqrt{\mu_r\varepsilon_r}$. In normal materials, as $\mu' >> \mu''$ and $\varepsilon' >> \varepsilon''$ in the application frequency range $(\psi_2 \approx 0)$, we obtained:

$$d = \frac{(2k+1)c}{4f\sqrt{|\mu_r\varepsilon_r|}}. \tag{7}$$

As this equation has the same form as the quarter-wavelength model, the derivation of this equation, which is an exact solution to the matching condition of peak frequencies, provides strict proof of the quarter-wavelength model.

2.3.2. $\psi_1 \neq 0$ but small. In some materials, the phase difference between $\vec{E}$ and $\vec{H}$ caused by $\sqrt{\mu_r/\varepsilon_r}$ cannot be ignored but it is also not too large ($\psi_1 \neq 0$ but small). Then, equation (4*c*) gives

$$f(2\beta d) = \frac{\sin(2\beta d)}{\sinh(2\beta d \cdot \tan\psi_2)} = \tan\psi_1. \tag{8a}$$

As ψ_1 is small, using a series expansion near $2\beta d = (2k+1)\pi$, function $f(2\beta d)$ can be written as:

$$f(2\beta d) = 0 - \frac{2\beta - (2k+1)\pi}{\sinh[(2k+1)\pi \cdot \tan\psi_2]} + \ldots \tag{8b}$$

Neglecting higher-order terms and using further approximation of the hyperbolic sine function, one gets $2\beta d \approx (2k+1)\pi(1-\delta)$ when k is small. Hence,

$$d_{rev} = \frac{(2k+1)(1-\delta)c}{4f\sqrt{|\mu_r\varepsilon_r|}\cos\psi_2} \tag{9}$$

where $\delta = \tan\psi_1 \tan\psi_2$. This equation is the revised version of equation (7); when $\mu' >> \mu''$ and $\varepsilon' >> \varepsilon''$ $(\psi_1 \approx \psi_2 \approx 0)$, it is totally equal to the quarter-wavelength model. Hence, this equation has a mechanism similar to the quarter-wavelength model, so we termed it the revised quarter-wavelength formula. Both the equations were derived from the phase matching condition which is an exact solution to the matching condition of peak frequencies. Hence, for microwave absorption materials with high permeability or permittivity, these equations are more accurate than Naito and Suetake's model $d = \frac{c}{2\pi f\mu''}$ which is an approximate solution for peak frequencies under $\frac{1}{4}\lambda_m/d >> \frac{\pi}{2}$ (λ_m is the wavelength of the medium).

3. Experimental comparison of the quarter-wavelength model and revised equation

For general microwave absorption materials, the dependence of the RL frequency on their electromagnetic parameters and thickness satisfies the quarter-wavelength model (equation (7)) [1, 8, 12, 18–20]. Figures 1(a) and (b) give an example of this scenario. Figure 1(a) displays the RL–f curves at various absorber thicknesses calculated using equations (1) and (2) for the MoS_2/Fe@Fe_3O_4 composite. The electromagnetic parameters, on which the RL–f curves are based, were taken from our previous work [28]. For a given absorber thickness, there is an RL peak appearing (or two RL peaks for higher thickness) in the 1–18 GHz range. The RL peak is sensitive to the absorber thickness and shifts to a lower frequency with increasing absorber thickness. We identified each peak frequency and its matching thickness (d_{exp}) and plotted them; they are denoted by green dots in figure 1(b). Figure 1(b) also shows the d_{cal}–f curve (black line) fitted using the quarter-wavelength model with $k = 0$. It is obvious that all the green dots (henceforth termed experimental dots for convenience) are located on the d_{cal}–f curves in the entire frequency range; hence, the RL peak frequency can be well

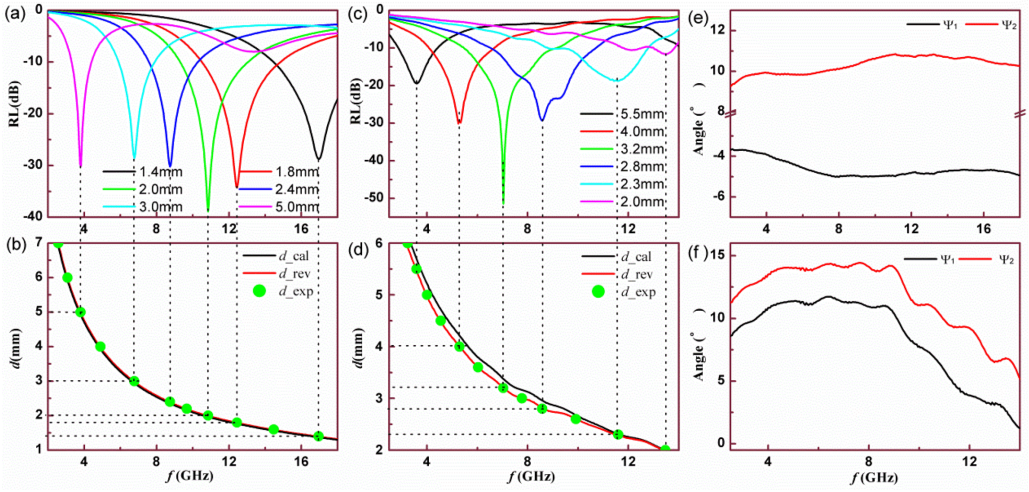


Figure 1. (a) Thickness-dependent RL–f curves of the MoS_2/Fe@Fe_3O_4 composite. (b) The d–f curves with $k = 0$ calculated by equation (7) (d_{cal}–f curve) and by equation (9) (d_{rev}–f curve) of the MoS_2/Fe@Fe_3O_4 composite. Each green dot (d_{exp}) represents a peak frequency and its matching thickness identified from the RL–f curve. (c) Thickness-dependent RL-f curves of the BN/GO/Fe_3O_4 composite. (d) The d–f curves with $k = 0$ calculated by equation (7) (d_{cal}–f curve) and by equation (9) (d_{rev}–f curve) of the BN/GO/Fe_3O_4 composite. (e), (f) The phase angle ψ_1 for $\sqrt{\mu_r/\varepsilon_r}$, as well as ψ_2 for $\sqrt{\mu_r\varepsilon_r}$, of the MoS_2/Fe@Fe_3O_4 composite and of the BN/GO/Fe_3O_4 composite, respectively.

described by the quarter-wavelength model. In addition, the d_{rev}–f curve (red line) fitted by the revised quarter-wavelength formula overlaps with the d_{cal}–f curve, implying that the two formulas are equivalent in this situation. For a material with RL peaks similar to these, the quarter-wavelength model will be feasible.

However, as mentioned above, in some microwave absorption materials, the dependence of the RL peak frequency on thickness deviates from the d_{cal}–f curve. Figures 1(c)–(d) show this deviation, based on the complex permittivity and permeability of the BN/GO/Fe_3O_4 composite measured in our previous work [25]. Clearly, the experimental dots locate lower than the d_{cal}–f curve, thus showing an observable deviation from the fitting curve. To determine whether this deviation could be described using our revised formula, we fitted the d_{rev}–f curve for the BN/GO/Fe_3O_4 composite using equation (9); this curve is indicated in red in figure 1(d). Obviously, this d_{rev}–f curve coincides with the experimental dots, which indicates that the revised quarter-wavelength formula is suitable for these types of composites.

To investigate the possible reason for the above-mentioned deviation, we calculated the phase angle ψ_1, as well as ψ_2, for the above two composites. The value of ψ_1 for the MoS_2/Fe@Fe_3O_4 composite (shown in figure 1(e)) is always small, while that for the BN/GO/Fe_3O_4 composite (shown in figure 1(f)) is larger in the range 4–9 GHz, where a notable deviation is observed. Considering the conditions under which the two equations (equations (7) and (9)) were derived, we attributed this deviation to the large absolute value of the phase angle of $\sqrt{\mu_r/\varepsilon_r}$ (that is $|\psi_1|$) .

To obtain further information in this regard and identify the commonly applied ranges of the two equations, we calculated the related microwave absorption properties for various microwave absorption materials.

Firstly, we consider whether the materials can be well described by the quarter-wavelength model. Here, we list the microwave absorption properties of two materials prepared and measured by our group: figure 2 for the Fe_3O_4/paraffin composite, and figure 3 for the graphene/paraffin composite. As observed in figures 2(c) and 3(c), all the experimental dots are located on the d_{cal}–f curves in the entire frequency range for both these materials. As for the value of ψ_1, one can see that its absolute value for both materials is always lower than 10°. For example, as shown in figure 2(b), the largest value of $|\psi_1|$ for the Fe_3O_4/paraffin composite is 7.3°; and the value of $|\psi_1|$ at its central frequency (3 GHz) is 4.9°. For the graphene/paraffin composite, as shown in figure 3(b), the corresponding values are 8.1° and 5.7°, respectively. Various samples with different materials, different volume fractions and different thicknesses were evaluated, and similar results were obtained irrespective of whether the magnetic or dielectric absorbent was present. In summary, generally when the absolute value of the phase angle ψ_1 is smaller than 10°, the dependence of the RL peak frequency on the electromagnetic parameters and thicknesses can be described by the quarter-wavelength model (equation (7)); moreover, the quarter-wavelength model and the revised quarter-wavelength formula are equivalent in this situation.

Then, we evaluated the materials showing the above-mentioned deviation, which is observed mostly in magnetic absorbents. Figure 4 shows the microwave absorption properties of a flaky carbonyl iron particle/polyurethane (mcip/polyurethane) composite. In this composite, as shown in figure 4(c), deviations between the experimental dots and the d_{cal}–f curves (black line) are observed after 1.5 GHz, especially when $k = 1$; moreover, the level of deviation increases with frequency in the shown range. At 1.5 GHz, as shown in figure 4(b), ψ_1 reaches 10° and then gradually increases with

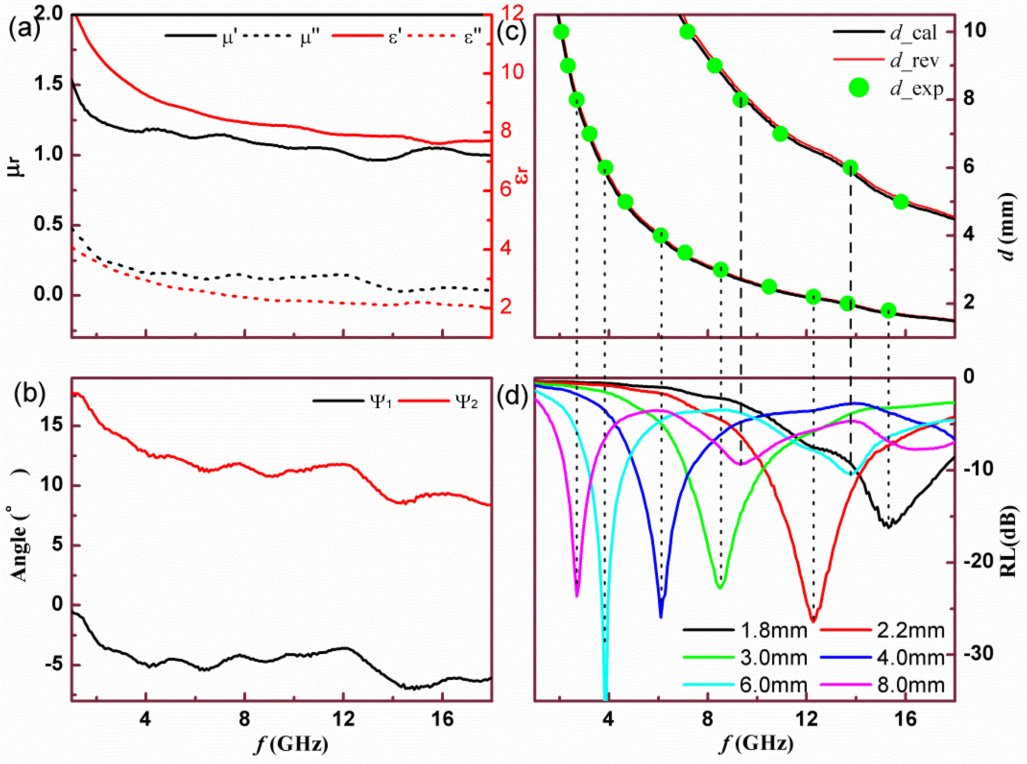


Figure 2. The characteristics of the Fe_3O_4/paraffin composite. (a) Complex permeability and permittivity. (b) The phase angle ψ_1 and ψ_2. (c) The d_{cal}–f curves calculated by equation (7) and d_{rev}–f curves calculated by equation (9) with $k = 0$, 1. (d) Thickness-dependent RL–f curves.

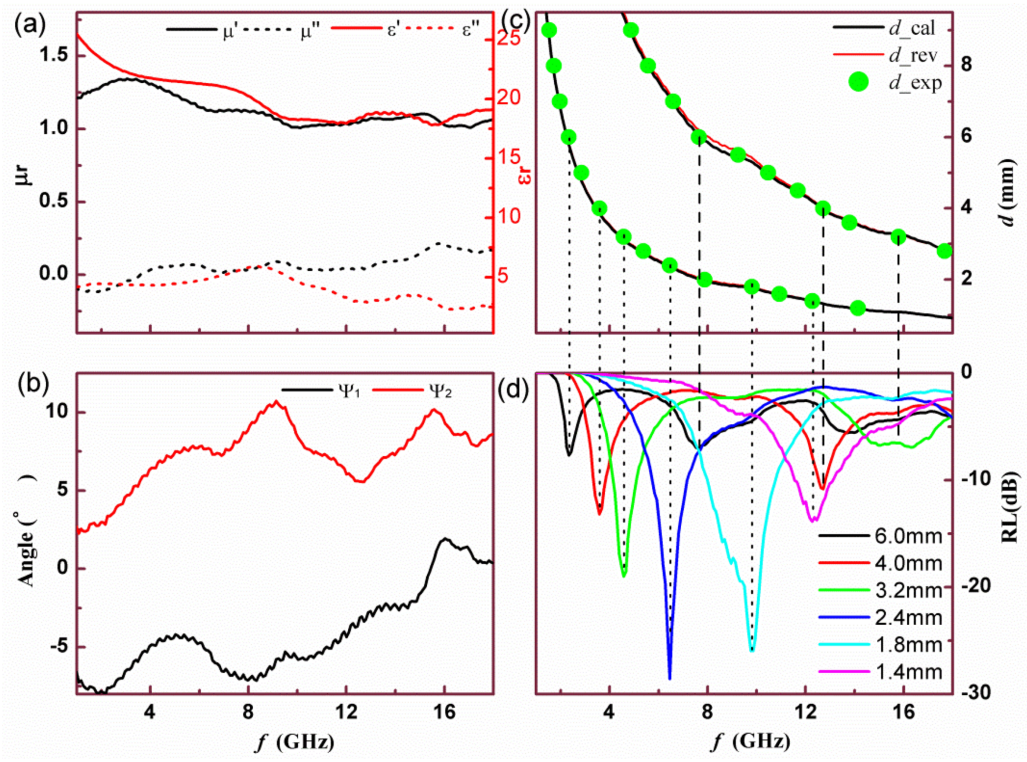


Figure 3. The characteristics of the graphene/paraffin composite. (a) Complex permeability and permittivity. (b) The phase angle ψ_1 and ψ_2. (c) The d_{cal}–f curves calculated by equation (7) and d_{rev}–f curves calculated by equation (9) with $k = 0$, 1. (d) Thickness-dependent RL–f curves.

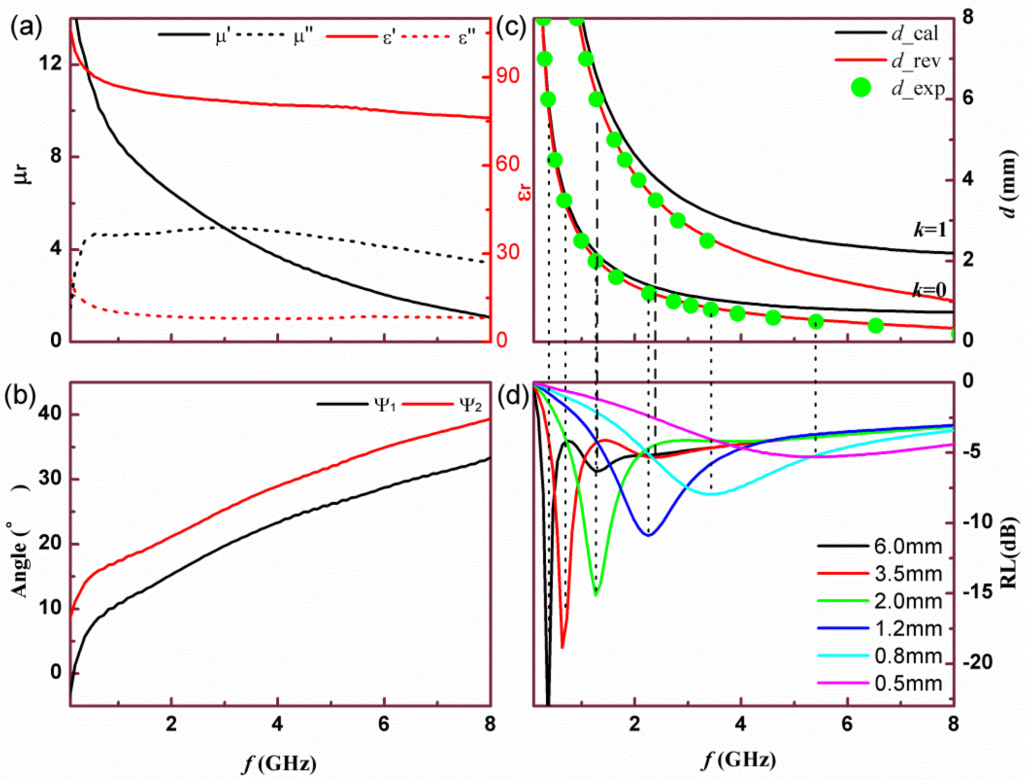


Figure 4. The characteristics of the mcip/polyurethane composite. (a) Complex permeability and permittivity. (b) The phase angle ψ_1 and ψ_2. (c) The d_{cal}–f curves calculated by equation (7) and d_{rev}–f curves calculated by equation (9) with $k = 0, 1$. (d) Thickness-dependent RL–f curves.

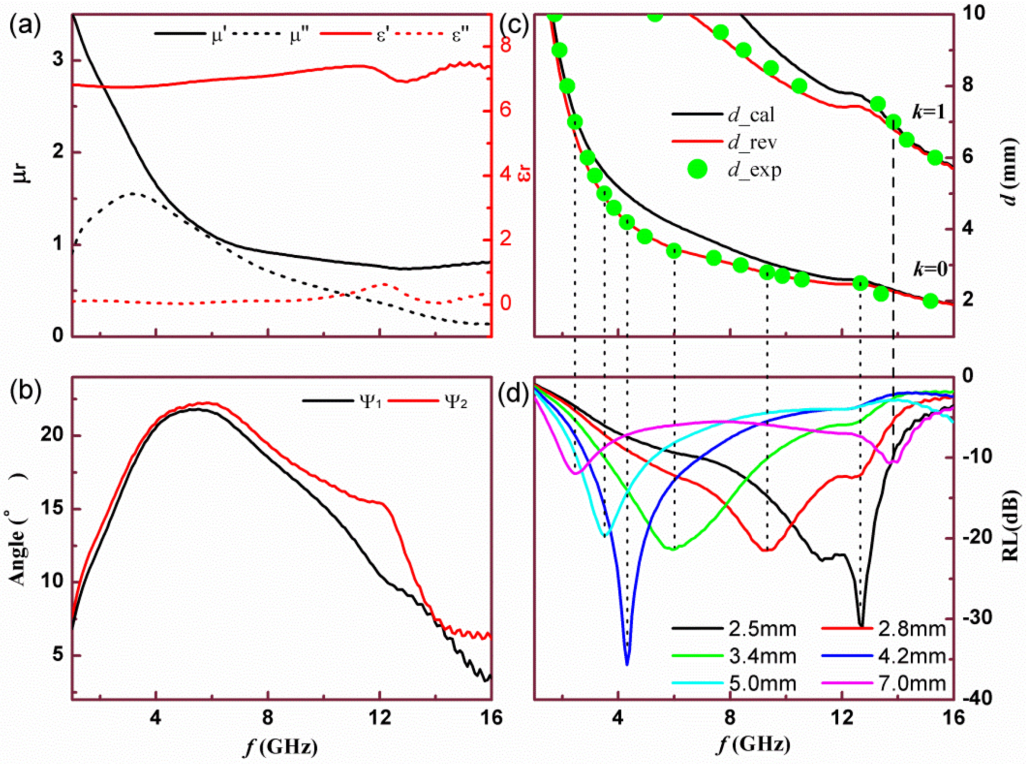


Figure 5. The characteristics of the Co_2Z/paraffin composite. (a) Complex permeability and permittivity. (b) The phase angle ψ_1 and ψ_2. (c) The d_{cal}–f curves calculated by equation (7) and d_{rev}–f curves calculated by equation (9) with $k = 0, 1$. (d) Thickness-dependent RL–f curves.

the increasing deviation. This observation supports our above argument. Considering the existence of deviation, we tried to fit the experimental dots using the revised quarter-wavelength formula (equation (9)), and show the results with red curves in figure 4(c). Clearly, the d_{rev}–f curves match well with the experimental dots; hence, the revised formula can be used to describe the dependence of the RL peak frequency on the electromagnetic parameters and thickness. As the deviation is far away from the central frequency (0.4 GHz) where the deviation can be ignored, the quarter-wavelength model can roughly be used [9].

It is worth noting that in some absorbers, the deviations are distinct and appear at the central frequency, and hence cannot be overlooked. Figure 5 shows the microwave absorption properties of a Co_2Z/paraffin composite (see [29] for preparation and measurement methods). Clearly, significant deviations between the experimental dots and d_{cal}–f curves are observed in 3–9 GHz, and they are highest around the central frequency (4.5 GHz). The trend of the deviation in the Co_2Z/paraffin composite is different from that of the mcip/polyurethane composite, but the regularity between the deviation and phase angle is similar. The deviation still appears in the frequency range with $|\psi_1|$ larger than 10°; in addition, the largest deviation is observed in the frequency range with the largest $|\psi_1|$. Further, although the quarter-wavelength model is not accurate enough to describe the experimental dots, the d_{rev}–f curves (red lines) can perfectly fit them. The related microwave absorption properties of some other materials with distinct deviations were also evaluated [30–33], and similar results were obtained. We also try to use the revised quarter-wavelength formula in situations involving more than one peak in an RL–f curve and with one peak frequency mismatching the quarter-wavelength model (similar to the RL curve with $d = 3.5$ mm shown in figure 4(d)) [24]. It was found that these peak frequencies mismatching the quarter-wavelength model could be described using the revised formula. As the revised quarter-wavelength formula is just a revised version of the quarter-wavelength model, we consider it inappropriate to evaluate the different RL peaks within the same thickness using two different models [24].

Therefore, we argue that if $|\psi_1| > 10°$ in a frequency range, a notable deviation will appear in this frequency range. In these materials, the revised quarter-wavelength formula can be used to describe the dependence of the RL frequency on their electromagnetic parameters and thickness. Further, there are few materials showing obvious RL peaks after $|\psi_1| = 30°$; hence we roughly set the upper limit of $|\psi_1|$ as 30° for the revised quarter-wavelength formula. Because notable deviations usually appear in microwave absorption materials with magnetic absorbents, if we consider the situation $\varepsilon' >> \varepsilon''$, the revised quarter-wavelength formula is applicable at least for $\mu'' \leq 1.73\mu'$.

Moreover, we identified a new method to reduce the absorber thickness using the notable deviation. It is known that a high $|\mu_r\varepsilon_r|$ is helpful in reducing the absorber thickness, which can be easily derived from equation (7). Here, comparing equation (9) with equation (7), we determined that appropriately increasing ψ_1 will be effective in reducing the absorber thickness. For instance, in the Co_2Z/paraffin composite (shown in figure 5), for the RL peak at 4.1 GHz, the lowest thickness is 4.2 mm; with $\psi_1 = 0$ and without changing $|\mu_r\varepsilon_r|$, the lowest thickness will be 5 mm. This method is particularly effective when the value of $|\mu_r\varepsilon_r|$ is high. Because a slight increase in ψ_1 will lead to a large decrease in thickness, the change of thickness could be enlarged by high $|\mu_r\varepsilon_r|$ according to equation (9). Hence, setting an appropriate value of ψ_1 is helpful in reducing absorber thickness.

4. Conclusion

Based on transmission line theory, we determined that the phase angle of Z_p should meet the phase matching condition $\psi = \psi_1 + \theta = 2n\pi$ once RL peaks appear. According to this condition, for microwave absorption materials with high permeability or permittivity, the dependence of the RL peak frequency on their electromagnetic parameters and thicknesses is deduced: $d = \frac{(2k+1)c}{4f\sqrt{|\mu_r\varepsilon_r|}}$ for $\psi_1 \approx 0$ (generally with $\mu' >> \mu''$ and $\varepsilon' >> \varepsilon''$); and $d_{rev} = \frac{(2k+1)(1-\delta)c}{4f\sqrt{|\mu_r\varepsilon_r|}\cos\psi_2}$ for $\psi_1 \neq 0$ but not too large. The experimental results demonstrated that when the phase angle (ψ_1) of $\sqrt{\mu_r/\varepsilon_r}$ is generally smaller than 10°, the former equation is perfect to fit the dependence of the RL peak frequency on electromagnetic parameters and thickness, where the two equations are identical. Above $|\psi_1| = 10°$ (generally, $|\psi_1| < 30°$), deviations between the experimental results and the fitting curves based on the former equation appear. In this situation, the latter equation can perfectly fit the dependence of the RL peak frequency on electromagnetic parameters and thickness.

However, as we did not evaluate a large number of materials, further studies are required to verify this revised formula. Additionally, this work is valid for a plane wave incident normally on an absorber backed by a perfect conductor; therefore, it needs to be improved for applications involving oblique incidence.

Despite these limitations, the model could accurately predict the matching thicknesses for absorption materials working at appointed frequencies or the RL peak frequencies for given absorbers. Moreover, this work demonstrated that the appropriate value of ψ_1 is effective in reducing absorber thickness; this is a new method to reduce absorber thickness. The findings of this work provide a theoretical basis for understanding and evaluating the properties of microwave absorption materials. Furthermore, they are useful in guiding research on the development and fabrication of microwave absorption materials for anti-electromagnetic radiation/interference applications and in the development of stealth radar technology.

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Appendix

The RL peaks appear when $|\Gamma|_f{}' = 0$ and $|\Gamma|_f{}'' > 0$. To discuss universal results, the influence of the change in $\mu_r\,(\varepsilon_r)$ with f is ruled out in the derivations, as $\mu_r\,(\varepsilon_r)$ changes differently with f in different absorbers. Therefore, with appropriate mathematical transformation, the derivatives can be written as:

$$|\Gamma|_f{}' = |\Gamma|_\psi{}' \cdot \frac{\partial \psi}{\partial f} = |\Gamma|_\psi{}' \cdot \cos^2\theta \cdot \frac{\partial \tan\theta}{\partial f} \qquad \text{(i)}$$

$$|\Gamma|_f{}'' = |\Gamma|_\psi{}'' \cdot \left(\frac{\partial \theta}{\partial f}\right)^2 > 0. \qquad \text{(ii)}$$

α and β always change with f and absorbers, hence, according to equation (4*c*), the value of $\tan\theta$ versus f in the first derivation is not zero. Similarly, $\cos\theta \neq 0$, as A_d and $\tan\theta$ have meanings. Therefore, the condition $|\Gamma|_f{}' = 0$ and $|\Gamma|_f{}'' > 0$ is equivalent to $|\Gamma|_\psi{}' = 0$ and $|\Gamma|_\psi{}'' > 0$. Clearly, this condition should be satisfied once an RL peak appears.

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